

BOURGOGNE AP, France

WMO: 072800

Lat: 47.2678N

Lon: 5.0883E

Elev: 221

StdP: 98.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-4.9	-10.3	1.6	-5.2	-8.0	2.0	-3.1	11.9	7.7	10.5	6.6	2.8	10	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	32.0	20.5	30.0	19.9	28.2	19.3	21.6	29.4	20.7	28.1	19.9	26.6	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.3	24.3	18.3	13.5	23.2	17.6	12.9	22.6	63.7	29.6	60.7	28.1	57.9	26.6	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.3	-10.3	34.7	3.0	2.1	-12.5	36.2	-14.2	37.4	-15.9	38.6	-18.1	40.2		
(5)			-10.6	23.3	3.0	0.8	-12.8	23.9	-14.6	24.3	-16.2	24.8	-18.4	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	2.6	3.7	7.4	10.5	14.3	18.3	20.4	19.9	16.1	11.8	6.3	3.2		
	DBStd	7.26	4.04	3.78	3.54	3.40	3.36	3.66	3.38	3.43	3.14	3.48	3.66	4.09		
	HDD10.0	896	232	177	94	35	4	0	0	0	1	22	118	212		
	HDD18.3	2802	489	409	339	236	130	46	18	22	82	202	360	470		
	CDD10.0	1352	1	1	14	49	138	250	322	306	182	78	8	1		
	CDD18.3	217	0	0	0	0	6	45	82	69	14	1	0	0		
	CDH23.3	2231	0	0	0	11	82	479	835	687	131	6	0	0		
	CDH26.7	748	0	0	0	0	9	150	306	258	24	0	0	0		
(14) Wind	WSAvg	3.3	3.5	3.8	3.8	3.6	3.2	3.2	3.2	2.9	3.1	3.0	3.1	3.4		
(15) Precipitation	PrecAvg	750	59	50	51	54	80	66	60	67	63	67	71	61		
	PrecMax	1079	126	117	164	149	196	176	138	139	265	178	259	171		
	PrecMin	476	15	6	5	2	6	0	6	11	9	6	6	0		
	PrecStd	137	28	27	33	37	38	37	30	34	47	38	43	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.2	20.2	24.8	27.9	33.5	34.7	35.9	29.3	24.1	17.4	13.5		
		MCWB	10.9	9.8	12.2	16.1	19.3	21.0	20.6	20.8	19.0	17.1	13.5	10.8		
	2%	DB	11.2	12.9	18.1	22.1	25.5	30.3	32.0	31.6	26.9	21.5	15.0	11.7		
		MCWB	9.3	9.1	11.3	14.1	17.6	20.3	20.4	20.4	18.3	16.5	12.3	10.1		
	5%	DB	9.8	11.1	15.8	19.9	23.7	28.2	30.0	29.5	24.6	19.3	13.3	10.4		
		MCWB	8.3	8.3	10.5	12.9	16.6	19.6	19.9	19.8	17.5	15.5	11.6	9.1		
	10%	DB	8.4	9.6	13.7	17.5	21.6	26.1	28.0	27.2	22.4	17.4	11.8	9.0		
		MCWB	7.2	7.5	9.4	11.9	15.5	18.5	19.1	19.0	16.4	14.4	10.3	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	10.6	13.4	16.9	20.4	22.7	22.4	22.5	20.5	18.4	14.1	11.5		
		MCDB	12.5	13.5	18.4	23.7	26.4	30.3	30.5	30.5	26.7	21.9	16.2	12.6		
	2%	WB	9.7	9.6	12.0	14.8	18.7	21.4	21.4	21.4	19.2	17.1	12.8	10.3		
		MCDB	10.8	12.1	16.5	20.8	24.4	28.4	29.7	29.2	25.1	20.3	14.4	11.4		
	5%	WB	8.4	8.6	10.9	13.5	17.3	20.2	20.6	20.5	18.1	15.9	11.7	9.3		
		MCDB	9.6	10.7	14.9	18.7	22.2	26.7	28.3	27.6	23.4	18.7	13.2	10.4		
	10%	WB	7.2	7.6	9.9	12.3	16.1	19.1	19.7	19.6	17.1	14.8	10.4	8.0		
		MCDB	8.3	9.4	13.2	16.6	20.4	25.0	26.5	25.8	21.2	17.0	11.5	8.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	7.5	9.6	10.5	10.7	11.5	11.7	11.5	10.4	8.9	6.5	5.4		
		MCDBR	6.8	10.5	13.7	14.9	14.3	14.7	15.0	15.3	13.9	11.9	8.6	6.9		
	5% WB	MCWBR	5.5	7.2	8.1	8.1	7.5	6.5	6.0	6.2	6.7	7.3	6.3	5.5		
		MCWBR	6.5	9.2	12.2	13.5	12.9	13.8	13.6	13.7	12.3	11.0	8.0	6.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.311	0.331	0.358	0.391	0.398	0.406	0.402	0.396	0.373	0.359	0.328	0.308		
		taud	2.460	2.409	2.347	2.259	2.287	2.297	2.306	2.339	2.384	2.418	2.458	2.456		
	Ebn at Noon	Ebn at Noon	777	832	860	861	867	859	857	847	835	786	746	733		
		Edh at Noon	66	85	105	126	127	127	124	115	101	84	66	60		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.04	1.87	3.14	4.41	5.16	6.00	5.73	4.96	3.83	2.35	1.24	0.85		
	RadStd	RadStd	0.14	0.30	0.42	0.61	0.56	0.56	0.49	0.43	0.32	0.18	0.12	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page